EduFin SQL Business Challenge

Phase 2 — Tuesday Submission

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STEP 2A — City-Level Loan Volume

```
WITH city_loans AS (
    SELECT
        dc.city_name AS city,
        COUNT(*) AS loans,
        COUNT(DISTINCT c.customer_id) AS customers,
        SUM(l.loan_amount) AS total_amount
    FROM loans l
    INNER JOIN customers c ON l.customer_id = c.customer_id
    INNER JOIN dim_city dc ON c.city_id = dc.city_id
    GROUP BY dc.city_name
)
SELECT
    city AS `City`,
    loans AS `Loans`,
    customers AS `Customers`,
    CONCAT(ROUND(total_amount * 100.0 / SUM(total_amount) OVER (), 2), '%') AS
`Portfolio Share (%)`
FROM city_loans
ORDER BY loans DESC;
```

```
+-----------+-----+---------+-------------------+
|       City|Loans|Customers|Portfolio Share (%)|
+-----------+-----+---------+-------------------+
|      Delhi|  209|      115|              4.01%|
|     Raipur|  207|      119|              4.15%|
|     Bhopal|  204|      118|              4.16%|
|Bhubaneswar|  199|      116|              4.13%|
|    Chennai|  191|      119|              3.73%|
|      Patna|  183|      114|              3.79%|
|    Lucknow|  179|      101|              3.67%|
|     Nagpur|  176|      110|              3.44%|
|   Guwahati|  176|       99|              3.56%|
|     Indore|  170|      104|              3.29%|
|     Ranchi|  168|      103|              3.27%|
|     Mumbai|  168|      106|              3.22%|
| Chandigarh|  168|       97|              3.46%|
|     Rajkot|  166|      106|              3.09%|
|      Kochi|  165|      105|              3.26%|
|   Dehradun|  163|       97|              3.16%|
|  Ahmedabad|  160|       97|              3.45%|
|     Jaipur|  160|       90|              3.21%|
|      Jammu|  159|       94|              3.19%|
|     Kanpur|  159|       95|              2.85%|
+-----------+-----+---------+-------------------+
only showing top 20 rows
```

STEP 2B — Default Count and Rate Per City

```
SELECT
    dc.city_name AS `City`,
    COUNT(*) AS `Loans`,
    SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS `Defaults`,
    CONCAT(
        ROUND(
            (SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100.0 /
COUNT(*)),
            2
        ),
        '%'
    ) AS `Default Rates (%)`
FROM loans l
INNER JOIN customers c ON l.customer_id = c.customer_id
INNER JOIN dim_city dc ON c.city_id = dc.city_id
GROUP BY dc.city_name
HAVING COUNT(*) >= 100
ORDER BY SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100.0 /
COUNT(*) DESC;
```

```
+----------+-----+--------+-----------------+
|      City|Loans|Defaults|Default Rates (%)|
+----------+-----+--------+-----------------+
|      Pune|  149|      30|           20.13%|
|    Ranchi|  168|      30|           17.86%|
|    Indore|  170|      28|           16.47%|
|    Jaipur|  160|      26|           16.25%|
|    Rajkot|  166|      26|           15.66%|
|     Kochi|  165|      25|           15.15%|
| Ahmedabad|  160|      24|           15.00%|
|     Jammu|  159|      23|           14.47%|
|    Nashik|  135|      19|           14.07%|
|   Lucknow|  179|      25|           13.97%|
|    Shimla|  144|      19|           13.19%|
|    Mumbai|  168|      22|           13.10%|
|    Nagpur|  176|      22|           12.50%|
|   Gangtok|  150|      17|           11.33%|
|    Bhopal|  204|      22|           10.78%|
|Chandigarh|  168|      18|           10.71%|
|     Delhi|  209|      22|           10.53%|
|    Raipur|  207|      21|           10.14%|
| Hyderabad|  159|      16|           10.06%|
|  Dehradun|  163|      16|            9.82%|
+----------+-----+--------+-----------------+
only showing top 20 rows
```

STEP 2C — Financial Exposure Per City

```
SELECT
    dc.city_name AS `City`,
    COUNT(*) AS `Loans`,
    CONCAT(ROUND(SUM(l.loan_amount) / 10000000.0, 2), ' Cr') AS `Portfolio Values`,
    CASE
        WHEN SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount * 1.0 ELSE 0
END) > 10000000.0
        THEN CONCAT(ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount
* 1.0 ELSE 0 END) / 10000000.0, 2), ' Cr')
        ELSE CONCAT(ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount
* 1.0 ELSE 0 END) / 100000.0, 2), ' L')
    END AS `Defaulted Values`,
    CONCAT(ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount * 1.0
ELSE 0 END) * 100.0 / SUM(l.loan_amount), 2), '%') AS `Loss Rates (%)`
FROM loans l
INNER JOIN customers c ON l.customer_id = c.customer_id
INNER JOIN dim_city dc ON c.city_id = dc.city_id
GROUP BY dc.city_name
ORDER BY SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount * 1.0 ELSE 0
END) * 100.0 / SUM(l.loan_amount) DESC;
```

```
+----------+-----+----------------+----------------+--------------+
|      City|Loans|Portfolio Values|Defaulted Values|Loss Rates (%)|
+----------+-----+----------------+----------------+--------------+
|    Ranchi|  168|         6.71 Cr|          1.4 Cr|        20.85%|
|      Pune|  149|         5.87 Cr|         1.14 Cr|        19.42%|
|    Mumbai|  168|         6.59 Cr|         1.06 Cr|        16.15%|
|    Jaipur|  160|         6.58 Cr|         1.03 Cr|        15.62%|
|    Rajkot|  166|         6.32 Cr|         97.93 L|        15.49%|
|    Indore|  170|         6.73 Cr|         1.04 Cr|        15.39%|
|     Jammu|  159|         6.52 Cr|         98.62 L|        15.12%|
|   Lucknow|  179|         7.51 Cr|         1.13 Cr|         15.1%|
|     Kochi|  165|         6.67 Cr|          1.0 Cr|        15.01%|
|    Nashik|  135|         5.62 Cr|         83.66 L|         14.9%|
| Ahmedabad|  160|         7.07 Cr|         1.05 Cr|        14.81%|
|    Shimla|  144|         5.99 Cr|         75.95 L|        12.67%|
|   Gangtok|  150|         6.08 Cr|         75.38 L|         12.4%|
|    Nagpur|  176|         7.06 Cr|         82.25 L|        11.66%|
| Hyderabad|  159|          6.7 Cr|         72.95 L|        10.89%|
|     Patna|  183|         7.77 Cr|         83.61 L|        10.76%|
|Chandigarh|  168|         7.09 Cr|          75.3 L|        10.61%|
|     Delhi|  209|         8.22 Cr|         83.12 L|        10.12%|
|    Bhopal|  204|         8.53 Cr|         84.25 L|         9.88%|
|  Dehradun|  163|         6.47 Cr|         63.55 L|         9.82%|
+----------+-----+----------------+----------------+--------------+
only showing top 20 rows
```

STEP 2D — Geographic Risk Pattern (State-Level Rollup)

```
SELECT
    dc.city_name AS `City`,
    COUNT(*) AS `Loans`,
    CONCAT(ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100 /
COUNT(*), 2), '%') AS `Default Rates (%)`,
    CASE
        WHEN ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100
/ COUNT(*), 2) > 15 THEN 'Critical'
        WHEN ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100
/ COUNT(*), 2) > 10 THEN 'High Risk'
        WHEN ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100
/ COUNT(*), 2) > 5 THEN 'Monitor'
        ELSE 'Safe'
    END AS `Risk Category`
FROM loans l
INNER JOIN customers c ON l.customer_id = c.customer_id
INNER JOIN dim_city dc ON c.city_id = dc.city_id
GROUP BY dc.city_name;
```

```
+-----------+-----+-----------------+-------------+
|       City|Loans|Default Rates (%)|Risk Category|
+-----------+-----+-----------------+-------------+
|  Bangalore|  148|            8.78%|      Monitor|
|      Kochi|  165|           15.15%|     Critical|
|Bhubaneswar|  199|            9.55%|      Monitor|
|      Jammu|  159|           14.47%|    High Risk|
|      Patna|  183|            9.29%|      Monitor|
|   Vadodara|  153|            7.19%|      Monitor|
|    Gangtok|  150|           11.33%|    High Risk|
|    Chennai|  191|            6.81%|      Monitor|
|    Lucknow|  179|           13.97%|    High Risk|
|     Shimla|  144|           13.19%|    High Risk|
|     Ranchi|  168|           17.86%|     Critical|
|     Mumbai|  168|            13.1%|    High Risk|
|  Ahmedabad|  160|            15.0%|    High Risk|
|    Kolkata|  129|            8.53%|      Monitor|
|   Dehradun|  163|            9.82%|      Monitor|
|       Pune|  149|           20.13%|     Critical|
|      Delhi|  209|           10.53%|    High Risk|
|     Raipur|  207|           10.14%|    High Risk|
| Chandigarh|  168|           10.71%|    High Risk|
|     Nagpur|  176|            12.5%|    High Risk|
+-----------+-----+-----------------+-------------+
only showing top 20 rows
```

STEP 2E — Regional Crisis Dashboard (Final Query)

```
WITH city_metrics AS (
    SELECT
        dc.city_name AS City,
        COUNT(*) AS `Loans`,
        COUNT(DISTINCT c.customer_id) AS `Customers`,
        SUM(l.loan_amount) AS total_portfolio,
        SUM(CASE
            WHEN l.loan_status = 'Defaulted'
            THEN l.loan_amount * 1.0
            ELSE 0
        END) AS default_exposure,
        ROUND(SUM(CASE
            WHEN l.loan_status = 'Defaulted' THEN 1
            ELSE 0
        END) * 100.0 / COUNT(*), 2) AS default_rate
    FROM loans l
    INNER JOIN customers c ON l.customer_id = c.customer_id
    INNER JOIN dim_city dc ON c.city_id = dc.city_id
    GROUP BY dc.city_name
    HAVING COUNT(*) >= 100
)

SELECT
    City,
    Loans,
    Customers,
    CONCAT(ROUND(total_portfolio / 10000000.0, 2), ' Cr') AS `Portfolio Value`,
    CASE
        WHEN default_exposure > 10000000
        THEN CONCAT(ROUND(default_exposure / 10000000.0, 2), ' Cr')
        ELSE CONCAT(ROUND(default_exposure / 100000.0, 2), ' L')
    END AS `Defaulted Value`,
```

```
    CONCAT(ROUND(default_rate, 2), '%') AS `Default Rates (%)`,
    ROUND((default_exposure / 10000000.0) * default_rate, 2) AS `Exposure Score`,
    RANK() OVER (
        ORDER BY (default_exposure / 10000000.0) * default_rate DESC
    ) AS `Exposure Rank`,
    CASE NTILE(4) OVER (ORDER BY default_exposure DESC)
        WHEN 1 THEN 'Critical'
        WHEN 2 THEN 'High'
        WHEN 3 THEN 'Moderate'
        ELSE 'Low'
    END AS `Risk Quartile`
FROM city_metrics
ORDER BY default_exposure DESC
LIMIT 15;
```

City	Loans	Customers	Portfolio Value	Defaulted Value	Default Rates (%)	Exposure Score	Exposure Rank	Risk Quartile
Ranchi	168	103	6.71 Cr	1.4 Cr	17.86%	24.97	1	Critical
Pune	149	94	5.87 Cr	1.14 Cr	20.13%	22.95	2	Critical
Lucknow	179	101	7.51 Cr	1.13 Cr	13.97%	15.85	5	Critical
Mumbai	168	106	6.59 Cr	1.06 Cr	13.10%	13.94	10	Critical
Ahmedabad	160	97	7.07 Cr	1.05 Cr	15.00%	15.71	6	Critical
Indore	170	104	6.73 Cr	1.04 Cr	16.47%	17.07	3	Critical
Jaipur	160	90	6.58 Cr	1.03 Cr	16.25%	16.71	4	Critical
Kochi	165	105	6.67 Cr	1.0 Cr	15.15%	15.17	8	Critical
Jammu	159	94	6.52 Cr	98.62 L	14.47%	14.27	9	High
Rajkot	166	106	6.32 Cr	97.93 L	15.66%	15.34	7	High
Bhopal	204	118	8.53 Cr	84.25 L	10.78%	9.08	14	High
Nashik	135	86	5.62 Cr	83.66 L	14.07%	11.77	11	High
Patna	183	114	7.77 Cr	83.61 L	9.29%	7.77	19	High
Delhi	209	115	8.22 Cr	83.12 L	10.53%	8.75	15	High
Nagpur	176	110	7.06 Cr	82.25 L	12.50%	10.28	12	High

The 3 cities to halt lending it immediately are Ranchi, Pune and Lucknow due to their high defaulted value exposure.

Custom Column

```
RANK() OVER (
   ORDER BY (default_exposure / 10000000.0) * default_rate DESC
) AS `Exposure Rank`

CASE NTILE(4) OVER (ORDER BY default_exposure DESC)
    WHEN 1 THEN 'Critical'
    WHEN 2 THEN 'High'
    WHEN 3 THEN 'Moderate'
    ELSE 'Low'
END AS `Risk Quartile`
```

> NTILE is better than hardcoded CASE thresholds because it automatically splits the cities into equal sized groups (quartiles) and also automatically adapts when new cities are added. Also since our portfolio data can grow or shrink in the near future, the hardcoded thresholds may become outdated making the user update the query time and again. However, in cases where you must stay consistent due to company policies, it is mandatory to use CASE thresholds as they can strictly follow absolute business rules.

WHY Gate

1. If you sorted your final list by default rate instead of exposure_rank — does your top-3 halt recommendation change? Name the specific cities that move in or out, and explain what that difference means for the CEO's decision.

> The top 3 halt recommendation changes from ('Ranchi', 'Pune', 'Lucknow') using total losses (defaulted amount) to ('Pune', 'Ranchi', 'Indore') using default_rank. Lucknow exits the halt list despite being the third-largest source of losses (1.13 Cr), while Indore enters despite lower absolute exposure (1.04 Cr). This means the CEO would halt cities based on frequency (rate) rather than financial damage (exposure), continuing to lend in a city that has already accumulated 1.13 Crores in losses. For a halt decision, total loss rank is correct because the CEO asked "where are we bleeding money" — not "which cities have high rates."

2. You excluded cities with fewer than 100 loans. Name one city that was excluded and explain: why does excluding it protect the business from making a bad decision?

> No, city was excluded as there were no cities with fewer than 100 loans.

3. Write your 3-city halt decision memo. Format: City Name → default rate → defaulted exposure → specific reason to halt THIS city and not the next one on the list.

> - 'Ranchi' → 17.86% → 1.4 Cr → It is the highest financial exposure in the portfolio (1.40 Cr), combined with one of the highest default rates. - 'Pune' → 20.13% → 1.14 Cr → It has the highest default rate (20.13%) from our entire list, while also having a decent financial exposure, making it the city with the worst failure frequency. - 'Lucknow' → 13.97% → 1.13 Cr → It has the 3rd highest financial exposure of 1.13 Cr

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